



SIMATS
ENGINEERING



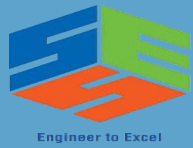
SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

From Caesar to Vigenère: Evolution of Substitution Ciphers

CSA5101-Cryptography and Network Security

Guided By:
E Anbalagan

Presented by:
N.Akshaya
192425129



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Introduction to Cryptography

What is Cryptography?

Cryptography is the technique of protecting information.

It converts **plaintext** into **ciphertext**.

The original message can be recovered using **decryption**.

It provides confidentiality and security.

Basic process:

Plaintext → Encryption → Ciphertext → Decryption → Plaintext

Example:

Plaintext: HELLO

Ciphertext: KHOOR

Substitution Cipher

What is a Substitution Cipher?

A substitution cipher replaces each letter with another letter according to a specific rule.

Example:

If:

A → D

B → E

C → F

Then:

CAT → FDW

Types discussed:

Caesar Cipher

Vigenère Cipher

Evolution:

Simple substitution → More complex substitution

Caesar Cipher

The Caesar cipher shifts every letter by the same number of positions.

Formula:

Encryption:

$$C = (P + K) \bmod 26$$

Decryption:

$$P = (C - K) \bmod 26$$

Where:

- P = Plaintext
- C = Ciphertext
- K = Key

Example:

Key = 3

Plaintext	H	E	L	L	O
Ciphertext	K	H	O	O	R

Therefore:

HELLO → KHOOR

Weakness of Caesar Cipher

Why is Caesar Cipher Weak?

- It has only **26 possible keys**.
- The same shift is used for every letter.
- It is vulnerable to **brute-force attacks**.
- Letter frequency can reveal the original message.

Example:

If the key is unknown, an attacker can try:

Key 1 → Key 2 → Key 3 → ... → Key 25

Problem:

HELLO → KHOOR

An attacker can easily test all possible shifts.

Vigenère Cipher

Vigenère Cipher

The Vigenère cipher improves Caesar cipher by using a **keyword**.
Instead of one fixed shift, it uses **different shifts** based on the letters of the key.

Formula:

$$C_i = (P_i + K_i) \bmod 26$$

Example:

Plaintext: HELLO

Key: KEY

Repeat the key:

KEYKE

Then different shifts are applied to each letter.

Result:

HELLO → RIJVS

Caesar vs Vigenère

Comparison

Feature	Caesar Cipher	Vigenère Cipher
Key	Single number	Keyword
Shift	Same shift	Different shifts
Security	Low	Higher
Keys	26 possible shifts	Many possible combinations
Frequency attack	Easier	More difficult
Complexity	Simple	More complex

Experimental Design & Implementation

Objective:

To compare the encryption of **Caesar** and **Vigenère** ciphers.

Input:

Message: HELLO

Caesar:

Key = 3

HELLO → KHOOR

Vigenère:

Key = KEY

HELLO → RIJVS

Tools Used:

- **Python**
- **Jupyter Notebook / Google Colab**
- **Basic mathematical operations**

Results & Analysis

Cipher	Plaintext	Key	Ciphertext
Caesar	HELLO	3	KHOOR
Vigenère	HELLO	KEY	RIJVS

Analysis

- Caesar uses one fixed shift.
- Vigenère uses multiple shifts.
- Vigenère produces a more varied ciphertext.
- Vigenère is more difficult to break using simple frequency analysis.
- Therefore, **Vigenère provides better security than Caesar.**

Conclusion

- Caesar cipher is a simple substitution cipher.
- Its fixed shift makes it easy to attack.
- Vigenère improves security using a repeated keyword.
- It applies multiple substitution shifts.
- This shows the **evolution from simple to more complex encryption.**



THANK YOU